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In [1]: import numpy as np
import pandas as pd

# Create the dataset
data = {
    'Country': ['France', 'Spain', 'Germany', 'Spain', 'Germany',
                'France', 'Spain', 'France', np.nan, 'France'],
    'Age': [44, 27, 30, 38, 40, 35, np.nan, 48, 50, 37],
    'Salary': [72000, 48000, 54000, 61000, np.nan,
               58000, 52000, 79000, 83000, 67000],
    'Purchased': ['No', 'Yes', 'No', 'No', 'Yes',
                  'Yes', 'No', 'Yes', 'No', 'Yes']
}

df = pd.DataFrame(data)

# Display original dataset
print("Original Dataset:")
print(df)

# Display dataset information
print("\nDataset Information:")
df.info()

# Handle missing values
df['Country'] = df['Country'].fillna(df['Country'].mode()[0])
df['Age'] = df['Age'].fillna(df['Age'].median())
df['Salary'] = df['Salary'].fillna(round(df['Salary'].mean()))

# Display dataset after handling missing values
print("\nDataset After Handling Missing Values:")
print(df)

# Convert categorical Country column into dummy variables
country_dummies = pd.get_dummies(df['Country'])

# Combine dummy variables with other columns
updated_dataset = pd.concat(
    [country_dummies, df[['Age', 'Salary', 'Purchased']]],
    axis=1
)

# Encode Purchased column: No = 0, Yes = 1
updated_dataset['Purchased'] = updated_dataset['Purchased'].map({
    'No': 0,
    'Yes': 1
})

# Display final preprocessed dataset
print("\nFinal Preprocessed Dataset:")
print(updated_dataset)

# Check for missing values
print("\nMissing Values in Final Dataset:")

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print(updated_dataset.isnull().sum())

# Display final dataset information
print("\nFinal Dataset Information:")
updated_dataset.info()
```

Original Dataset:

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	NaN	50.0	83000.0	No
9	France	37.0	67000.0	Yes

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 10 entries, 0 to 9

Data columns (total 4 columns):

#	Column	Non-Null Count	Dtype
0	Country	9 non-null	str
1	Age	9 non-null	float64
2	Salary	9 non-null	float64
3	Purchased	10 non-null	str

dtypes: float64(2), str(2)

memory usage: 532.0 bytes

Dataset After Handling Missing Values:

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	63778.0	Yes
5	France	35.0	58000.0	Yes
6	Spain	38.0	52000.0	No
7	France	48.0	79000.0	Yes
8	France	50.0	83000.0	No
9	France	37.0	67000.0	Yes

Final Preprocessed Dataset:

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	0
1	False	False	True	27.0	48000.0	1
2	False	True	False	30.0	54000.0	0
3	False	False	True	38.0	61000.0	0
4	False	True	False	40.0	63778.0	1
5	True	False	False	35.0	58000.0	1
6	False	False	True	38.0	52000.0	0
7	True	False	False	48.0	79000.0	1
8	True	False	False	50.0	83000.0	0
9	True	False	False	37.0	67000.0	1

Missing Values in Final Dataset:

France	0
Germany	0
Spain	0

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Age          0
Salary       0
Purchased    0
dtype: int64
```

Final Dataset Information:

```
<class 'pandas.DataFrame'>
```

```
RangeIndex: 10 entries, 0 to 9
```

```
Data columns (total 6 columns):
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#	Column	Non-Null Count	Dtype
0	France	10 non-null	bool
1	Germany	10 non-null	bool
2	Spain	10 non-null	bool
3	Age	10 non-null	float64
4	Salary	10 non-null	float64
5	Purchased	10 non-null	int64

```
dtypes: bool(3), float64(2), int64(1)
```

```
memory usage: 402.0 bytes
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In []: